BRIEF STATE RUMINATION INVENTORY (BSRI)

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Abstract

The Brief State Rumination Inventory (BSRI) was developed as a measure to capture state rumination, complementing existing measures of trait rumination. Drawing from a pool of items derived from well-established trait questionnaires, this self-report instrument was crafted to encompass both the Dutch and English language and suitable for both paper-and-pencil and online formats. A comprehensive multi-group confirmatory factor analysis across three distinct samples yielded a coherent unidimensional structure with eight items. Rigorous validation procedures, including assessments of face, concurrent, divergent, and criterion validity further confirmed the robustness of its psychometric properties. Subsequent validation efforts across multiple languages underscored its reliability. The BSRI is a valid and reliable self-report questionnaire for capturing state rumination.

Keywords

Brief State Rumination Inventory, state rumination, emotion regulation

Introduction

Depressive rumination is characterized as a maladaptive type of self-focus (Mor & Winquist, 2022), entailing persistent attention towards the causes and implications of one's negative emotional state (Nolen-Hoeksema et al., 2008). Rumination is associated with substantial adverse effects, such as heightened susceptibility to experiencing depressive episodes (Nolen-Hoeksema, 2000) and prolonged periods of

depression (Kuehner & Weber, 1999). Moreover, it is also implicated in the development and exacerbation of other mental disorders, including anxiety disorders, substance abuse issues, and bulimic symptoms (Nolen-Hoeksema et al., 2007).

Theoretical Foundations

Rumination has been conceptualized both as a state and a trait phenomenon, first as a transient response to negative mood (Martin & Tesser, 1996) and second as a recurring tendency involving repetitive and passive self-focus in reaction to feelings of depression (Nolen-Hoeksema et al., 2008). Solid research shows that trait rumination predicts several negative outcomes, including clinical depression (Nolen-Hoeksema, 2000), depressive recurrence (Michalak et al., 2011), and suicide (Rogers & Joiner, 2017). Various self-report measures evaluate trait rumination as a mechanism applicable across different disorders, with a primary emphasis on its repetitive and perseverative nature. These tools include the Ruminative Response Scale (RRS; Nolen-Hoeksema & Morrow, 1991), the Rumination-Reflection Questionnaire (RRQ; Trapnell & Campbell, 1999), the Repetitive Thinking Questionnaire (RTQ; Mahoney et al., 2012), and the Perseverative Thinking Questionnaire (PTQ; Ehring et al., 2011).

Alternatively, rumination can be perceived as a temporary cognitive reaction usually triggered by situational cues and perceived discrepancies between one's goals and current emotional state (e.g., Martin & Tesser, 1996; Watkins & Nolen-Hoeksema, 2014). The impact of state rumination has been explored in various laboratory studies, revealing detrimental effects on mood, cognitive processes, problem-solving abilities, goal-directed behavior, and psychopathological symptoms (Lyubomirsky et al., 2015). In line with this, significant distinctions between trait and state rumination have been observed. For example, state rumination predicts subsequent negative emotions, ineffective recovery from sadness, and heightened emotional reactivity to social stressors independently of trait rumination (Hilt et al., 2015; Moberly & Watkins, 2008). These findings collectively suggest that state rumination provides valuable insights into mental functioning that are not fully captured by trait rumination.

The growing interest in state rumination has led to the development of several measures for this phenomenon. However, existing measures focus on particular situations, such as post-offense rumination

(McCullough et al., 2007) or reflection after an intelligence test (LeMoult et al., 2013), which limits their applicability to broader contexts. Some studies have relied on experience sampling but lacked a rationale for clustering items or thorough validation (Robert et al., 2013), and they typically evaluate only internal reliability (i.e., Cronbach's α) while neglecting dimensionality (i.e., factor analysis). The Brief State Rumination Inventory (BSRI) was developed with the aim of providing researchers and clinicians with a brief, psychometrically robust self-report instrument suitable for both laboratory and real-world assessments.

Development and Initial Validation of the Scale

For the development of the BSRI, a pool of 13 items was used to capture maladaptive state rumination, which was defined as the momentary occurrence of thoughts that focus one's attention on distress as well as on its possible causes and implications (Nolen-Hoeksema et al., 2008). The development process involved a thorough examination of established measures of trait rumination and repetitive negative thinking, namely the RRS, the PTQ, the RTQ, and the RRQ. In practical terms, each rumination scale comprised items adaptable for a state questionnaire suitable for laboratory use, excluding those that required actions like "going someplace alone to think about your feelings" (e.g., RRS). Moreover, the items were translated into Dutch and then back-translated to English, aiming to provide a succinct, unidimensional measure of state rumination. Finally, to increase the measure's sensitivity and make it useful for repeated administration, each item is positioned on a 100-mm visual analogue scale (VAS) on a continuum from "completely disagree" to "completely agree."

The original validation of the instrument relied on three independent samples: two samples of Dutch-speaking Belgian volunteers (n = 155 and n = 141, respectively) who were administered the paper version of the instrument, and one English-speaking sample (n = 199) with online administration (i.e., Prolific Academic). Multi-group confirmatory factor analysis identified a unidimensional solution with eight items. Furthermore, it was confirmed that the BSRI showed weak (metric) invariance, indicating that item loadings may be deemed equal across various groups, i.e., the measurement model has the same structure and the same underlying constructs of interest and the scale (unit of measurement) of the latent

variable is identical across groups. Moreover, the questionnaire demonstrated partial strong (scalar) invariance, with equivalent item intercepts, as well as partial strict invariance, featuring equivalent item errors. According to current guidelines, it is suggested that under conditions of partial measurement invariance, latent factors can be meaningfully compared and interpreted across diverse groups (Vandenberg & Lance, 2000).

Validity

To assess the face validity of the BSRI, a small group of experts in rumination research (n = 9) were asked to evaluate the items. They confirmed that the items accurately reflected the construct of rumination as defined by Nolen-Hoeksema et al. (2008). Moreover, in two separate samples, the BSRI showed positive correlations with trait brooding (r = .60 and .49), depressive symptoms (r = .49 and .54), and negative affect (r = .54 and .52), while displaying negative correlations with positive affect (r = -.18 and -.23) and trait mindfulness (r = -.39 and -.31). Finally, the BSRI demonstrated high sensitivity (Cohen's d = 0.55) in detecting changes in state rumination among individuals subjected to a rumination induction procedure, compared to those exposed to a procedure designed to elicit self-related but non-ruminative thoughts.

Internal Reliability

In their original validation study, Marchetti et al. (2018) reported excellent levels of internal reliability, i.e. internal consistency (Cronbach's α = .89 and .91). Similarly, very high levels of internal consistency were reported across individuals who differed with respect to age, mental health status, and experimental context. For instance, Zhang et al. (2024) reported high levels of internal reliability in college students (Cronbach's α = .90), while Owens and Bunce (2023) reported similar levels in adolescents (Cronbach's α =.86 and .90). Several studies relied on the BSRI when measuring state rumination in both healthy adults (Niculescu et al., 2024, Cronbach's α = .90) and individuals with major depression and other comorbid mental disorders (Bø et al., 2024; Cronbach's α = .86). Finally, the BSRI was also adopted in EEG studies, showing high levels of internal consistency (Luo et al., 2024; Cronbach's α = .78 and .96). These findings underscore the BSRI's robust internal reliability.

Subsequent Evidence of Psychometric Properties

While the BSRI has been translated into various languages like Danish, French, German, Hebrew, Icelandic, Italian, Japanese, Korean, Spanish, and Swedish, only the Turkish (Altan-Atalay et al., 2020) and Chinese (Wang et al., 2022) versions underwent a comprehensive validation process. Consistent with the original version, the Turkish BSRI demonstrated a unidimensional latent structure in a moderate-sized sample (n = 192), exhibiting high internal consistency (Cronbach's α = .91) and excellent split-half reliability (r = .93). It also exhibited good convergent validity, showing significant correlations with several measures of trait rumination (ranging from r = .60 to r = .73), and divergent validity, displaying a negative correlation with positive affect (r = -.38). Furthermore, the BSRI's incremental validity was demonstrated by showing that it predicts depressive symptoms even after controlling for age, sex, and trait rumination measures. Lastly, the BSRI proved capable of capturing momentary fluctuations in rumination following a state rumination induction (Cohen's d = .76).

Wang et al. (2022) translated the BSRI into Chinese and subjected the instrument to a validation procedure. Both exploratory and confirmatory factor analysis confirmed that the eight items loaded on one factor, with moderate to strong inter-item correlations (ranging from r = .43 to r = .77), high internal reliability (Cronbach's α = .93), and excellent split-half reliability (r = .86). Furthermore, the BSRI total score was positively correlated with trait rumination, depressive symptoms, and negative affect. Finally, the instrument could detect changes in state rumination after rumination induction when compared to a distraction induction (Cohen's d = .64).

Instructions on Administration

The BRSI can be administered through both online surveys and paper-based formats (see Appendix for the items). Participants can receive the BSRI along with printed instructions, and researcher supervision during completion is unnecessary. Although research on the minimum interval for administering the BSRI is currently absent, it is common practice in laboratory experiments to conduct repeated administrations within a short timeframe, often as brief as 10 minutes or shorter.

How to Score, Analyze, and Present the Data

All items are assessed using a 100-mm visual analogue scale (VAS), ranging from "completely disagree" (0) to "completely agree" (100), with all items scored in a forward manner. The total BSRI score is derived by summing scores across all items. In the absence of current research, it is recommended to avoid imputing missing data for the BSRI. Utilizing incomplete responses may compromise the reliability of the measure.

Applications to meditation and mindful practice

Althought the BSRI has demonstrated a negative association with trait mindfulness (r = -.31; Marchetti et al., 2018), it has only occasionally been employed in research on in-vivo meditation and mindful practice. For instance, Owens et al. (2022) examined the impact of a 20-minute nature-based meditation on state rumination, as measured with the BSRI. This nature-based meditation started and ended with a brief grounding breathing exercise from mindfulness-based approaches designed to center individuals in the present moment and encourage non-judgmental attention, especially towards their natural environment through specific exercises. The analysis revealed that the participants in the nature-based meditation group experienced a statistically significant reduction in state rumination compared to those in the indoor meditation and control conditions. More recently, Shireen et al. (2024) explored whether state rumination could mediate the impact of body scan meditation, as opposed to a control condition, on negative affect among emotional eaters In body scan meditation, participants were instructed to intentionally direct their attention to different parts of the body and observe without judgment or reaction. If thoughts and emotions arose, they were guided to briefly acknowledge them and return their focus to the body. Although the BSRI exhibited excellent internal consistency, state rumination did not function as a statistically significant mediator. Although research is currently limited, the BSRI appears to be a promising tool for investigating multiple aspects of meditation and mindful practice.

Limitations

The BSRI has three limitations. First, the BSRI was initially devised to measure state rumination in adults and young adults, with its suitability for children and the elderly yet to be established in terms of

validity and reliability. Future research endeavors could prioritize expanding the application of this questionnaire across different age groups, encompassing the whole lifespan. Second, the current length of the BSRI, comprising eight items, may pose challenges for repeated administration throughout the day, particularly in contexts such as Experience Sampling Method (ESM) studies. There is a need for future investigations aimed at crafting a shorter version of the questionnaire and exploring whether this abbreviated questionnaire maintains comparable psychometric properties to the original. Such refinement holds the potential to enhance the instrument's suitability for a wider array of applications. Third, despite being translated into numerous languages, the BSRI's validation was conducted in only four languages (e.g., Dutch, English, Turkish, and Chinese), thereby constraining our understanding of its cross-cultural validity. Future research should empirically validate the BSRI across a more diverse spectrum of languages and cultural contexts, fostering a thorough comprehension of its applicability.

Summary

The BSRI is an 8-item self-report questionnaire for state rumination, which has sound psychometric properties. Its versatility is evident through widespread application across diverse research domains, spanning various age groups, mental health conditions, and experimental settings. Furthermore, its validation across different languages and cultures underscores its validity and reliability as a valuable tool for assessing state rumination.

Appendix: The Brief State Rumination Inventory

Instructions: Please respond to the following items by referring to the way you feel or think right now. For each item, please mark a vertical line on the horizontal line on the visual analogue scale to indicate the degree to which you agree or disagree with the statement.

1. Right now, I am reflecting about my mood.
2. Right now, I wonder why I react the way I do.
3. Right now, I wonder why I always feel the way I do.
4. Right now, I am thinking: “why do I have problems other people don't have?”.
5. Right now, I am rehashing in my mind recent things I've said or done.
6. Right now, I am thinking “why can't I handle things better?”.

7. Right now, it is hard for me to shut off negative thoughts about myself.
8. Right now, I wonder why I can't respond in a better way.

Note: All items are measured on a 100-mm visual analogue scale (VAS) ranging from "completely disagree" (0) to "completely agree" (100)

Scoring instructions: All items are forward scored. The BSRI total score is computed by summing all items.

The scale is freely available for research groups to use and should be cited as: Marchetti, I., Mor, N. Chiorri, C., & Koster, E. H. (2018). The Brief State Rumination Inventory (BSRI): Validation and psychometric evaluation. *Cognitive Therapy and Research, 42*, 447-460. https://doi.org/10.1007/s10608-018-9901-1.

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